TU Dortmund University
Department of Statistics

Analysis of Student Performance Based on Gender and Parental Education

Application Report

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Contents

1 Introduction

Understanding the factors that influence academic performance is a central topic of educational research. Previous studies have shown that both gender and family background play a role in shaping student outcomes. Identifying such patterns is important for designing targeted interventions and informing education policy.

This report investigates whether Mathematics and Language scores differ significantly across gender and parental education level. The analysis is based on a dataset of 486 students, each assessed in both subjects. The hypothesis test results showed that male students score significantly higher than female students in Mathematics ($t = 3.776$, $p < 0.001$), while female students score significantly higher in Language ($t = -6.409$, $p < 0.001$). Regarding parental education, scores differ significantly across education levels in both Mathematics ($F = 6.223$, $p < 0.001$) and Language ($F = 14.241$, $p < 0.001$), with students whose parents have a high school diploma scoring notably lower than other groups.

All analyses are carried out in Python, using the NumPy, pandas, SciPy, Matplotlib, and scikit-posthocs libraries. The remainder of this report is structured as follows. Section 2 describes the dataset and research questions. Section 3 presents the statistical methods employed. Section 4 contains the descriptive and inferential analyses. Finally, Section 5 summarises the findings and discusses their implications.

2 Problem Description and Data

2.1 Problem Description

The task is to investigate whether there are statistically significant differences in students' Math and Language scores with respect to gender and parental level of education. We are trying to address two questions:

1. Is there a statistically significant difference between male and female students in their Math and Language scores?

2. Is there a difference in scores between parental education levels? If so, which groups differ from each other?

2.2 Data

The dataset contains 972 observations for 486 students, each of whom was assessed in both Math and Language scores ranging from 0 to 100. Each student contributes two rows. In addition to the test scores, the dataset includes the student's gender (male or female) and the highest parental level of education in four categories: high school, associate's degree, bachelor's degree, and master's degree.

3 Methods

3.1 Descriptive Statistics

We summarise the data using the sample mean, median, and standard deviation. We use boxplots and histograms to visualise the distributions.

3.2 Two-Sample t-Test

Research Question 1 asks whether male and female students differ in their mean scores. Since gender has two categories, we need a test that compares two independent group means. The two-sample t-test is designed for this purpose. The hypotheses are

$$H_0 : \mu_1 = \mu_2, \tag{1}$$
$$H_1 : \mu_1 \neq \mu_2. \tag{2}$$

The test statistic under equal variances is

$$t = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}}, \tag{3}$$

with pooled standard deviation

$$s_p = \sqrt{\frac{(n_1 - 1)\, s_1^2 + (n_2 - 1)\, s_2^2}{n_1 + n_2 - 2}}. \tag{4}$$

Under H_0, t follows a Student's t-distribution with $\nu = n_1 + n_2 - 2$ degrees of freedom. We reject H_0 at significance level α if $p < \alpha$.

However, the t-test relies on two assumptions: (1) the observations in each group are approximately normally distributed, and (2) the two populations have equal variances (homoscedasticity). If these assumptions are violated, the p-values may be unreliable. We therefore verify them before applying the test, using the procedures described in Section 3.3.

3.3 Assumption Checks

3.3.1 Shapiro–Wilk Test for Normality

Both the t-test and ANOVA assume that the scores within each group follow an approximately normal distribution. To verify this, we apply the *Shapiro–Wilk test*. It tests

$$H_0 : \text{The sample comes from a normal distribution,} \tag{5}$$
$$H_1 : \text{The sample does not come from a normal distribution.} \tag{6}$$

If the Shapiro–Wilk test rejects normality ($p < 0.05$) for a group, we cannot rely on parametric tests for that group and must consider non-parametric alternatives (such as the Kruskal–Wallis test or the Mann–Whitney U test).

3.3.2 Levene's Test for Equality of Variances

The pooled t-test and one-way ANOVA also assume that all groups share the same variance. We check this with *Levene's test* (median-based variant), which tests

$$H_0 : \sigma_1^2 = \sigma_2^2 = \cdots = \sigma_k^2, \tag{7}$$
$$H_1 : \text{at least one variance differs.} \tag{8}$$

The median-based variant is chosen because it is resistant against departures from normality. If Levene's test rejects H_0, equal variances cannot be assumed and a Welch correction or a non-parametric test should be used instead.

3.4 One-Way Analysis of Variance (ANOVA)

Research Question 2 asks whether scores differ across four parental education levels. Since we now compare more than two groups, the t-test is no longer appropriate — performing multiple pairwise t-tests would inflate the Type I error rate. One-way ANOVA addresses this by testing all group means simultaneously. The hypotheses are

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k, \tag{9}$$
$$H_1 : \text{at least one } \mu_i \neq \mu_j. \tag{10}$$

The test statistic is

$$F = \frac{\text{MSB}}{\text{MSW}}, \tag{11}$$

where

$$\text{MSB} = \frac{\sum_{i=1}^{k} n_i \left(\bar{X}_{i\cdot} - \bar{X}_{\cdot\cdot}\right)^2}{k-1}, \qquad \text{MSW} = \frac{\sum_{i=1}^{k} \sum_{j=1}^{n_i} \left(X_{ij} - \bar{X}_{i\cdot}\right)^2}{N-k}. \tag{12}$$

Under H_0, F follows an F-distribution with $k-1$ and $N-k$ degrees of freedom. ANOVA requires the same assumptions as the t-test: normality within each group and homogeneity of variances.

Since the Shapiro–Wilk test rejects normality for some subgroups (see Section 4), we additionally apply the *Kruskal–Wallis test* as an additional check. This non-parametric alternative compares mean ranks rather than means and does not require the normality assumption, making it appropriate when the parametric assumptions are questionable.

3.5 Post-Hoc Pairwise Comparisons

If ANOVA or the Kruskal–Wallis test rejects H_0, we know that at least one group differs, but not *which* groups differ from each other. With $k = 4$ education levels there are $\binom{4}{2} = 6$ pairwise comparisons. Performing each at $\alpha = 0.05$ inflates the probability of at least one false positive (the family-wise error rate).

To control this, we apply two post-hoc procedures. First, *Tukey's HSD* compares all pairs of group means while controlling the family-wise error rate; it assumes equal variances (confirmed by Levene's test) and is reasonably resistant to small non-normality. Second, *Dunn's test* with a *Bonferroni correction* serves as a non-parametric alternative. It compares mean ranks between every pair and multiplies each p-value by the number of comparisons m to keep the overall error rate at most α. Dunn's test is appropriate when normality is questionable in some groups, as is the case here.

By reporting both tests we can assess whether the conclusions hold regardless of distributional assumptions.

4 Evaluation

4.1 Descriptive Analysis

The dataset contains 972 observations with no missing values across any variable. The sample comprises 245 female and 241 male students. The most common parental education level is associate's degree (36.4%), followed by high school (32.1%), bachelor's degree (19.3%), and master's degree (12.1%).

Table 1 presents the descriptive statistics broken down by subject and gender. In Mathematics, male students achieve a higher mean score ($\bar{x} = 68.95$, $s = 14.43$) than female students ($\bar{x} = 63.86$, $s = 15.26$). In Language on the other hand, female students score higher on average ($\bar{x} = 73.00$, $s = 14.01$) than male students ($\bar{x} = 64.93$, $s = 13.75$).

Table 1. Descriptive statistics of Mathematics and Language scores by gender.

Subject	Gender	n	Mean	Median	Std. Dev.
Math	Female	245	63.86	64.0	15.26
Math	Male	241	68.95	70.0	14.43
Language	Female	245	73.00	73.0	14.01
Language	Male	241	64.93	65.5	13.75

Table 2 shows the descriptive statistics by parental education level and subject. Students whose parents hold a master's degree achieve the highest mean scores in both Mathematics ($\bar{x} = 69.75$) and Language ($\bar{x} = 75.53$), whereas students with high-school-educated parents score lowest in both subjects Mathematics ($\bar{x} = 62.24$) and Language ($\bar{x} = 63.53$).

Table 2. Descriptive statistics of scores by parental education level and subject.

Parental Education	Subject	n	Mean	Median	Std. Dev.
Associate's degree	Math	177	67.81	66.0	15.24
Associate's degree	Language	177	70.02	72.0	14.21
Bachelor's degree	Math	94	68.44	67.0	15.05
Bachelor's degree	Language	94	72.06	73.0	14.27
High school	Math	156	62.24	63.0	14.04
High school	Language	156	63.53	65.0	13.40
Master's degree	Math	59	69.75	73.0	15.15
Master's degree	Language	59	75.53	76.0	13.55

Figure 1 shows the score distributions broken down by parental education level and subject. The high school group consistently occupies the lowest position in both subjects, while the master's degree group tends to score highest.

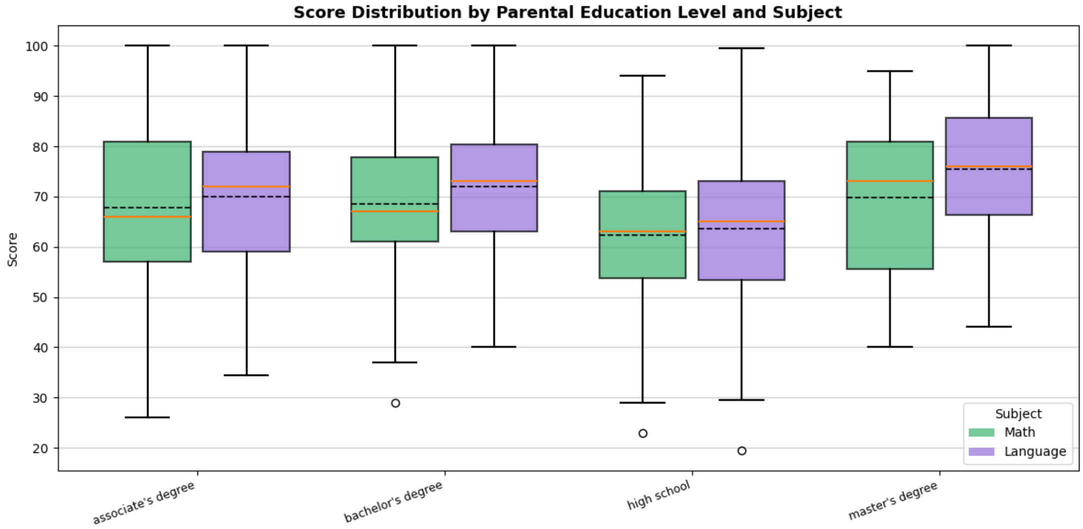

Figure 1. Score distributions by parental education level and subject.

4.2 Gender Differences in Scores (Research Question 1)

We test whether mean scores differ between male and female students, separately for each subject. The significance level is set at $\alpha = 0.05$ throughout.

4.2.1 Mathematic Scores

The Shapiro–Wilk test does not reject normality for either group (males: $W = 0.988$, $p = 0.050$; females: $W = 0.994$, $p = 0.501$). Levene's test confirms equal variances ($W = 0.006$, $p = 0.938$). Both assumptions are satisfied, so the two-sample t-test is applied.

The test yields $t = 3.776$ with 484 degrees of freedom and $p < 0.001$. We reject H_0, therefore male students score significantly higher than female students in Mathematics. Figure 2 visualises this difference.

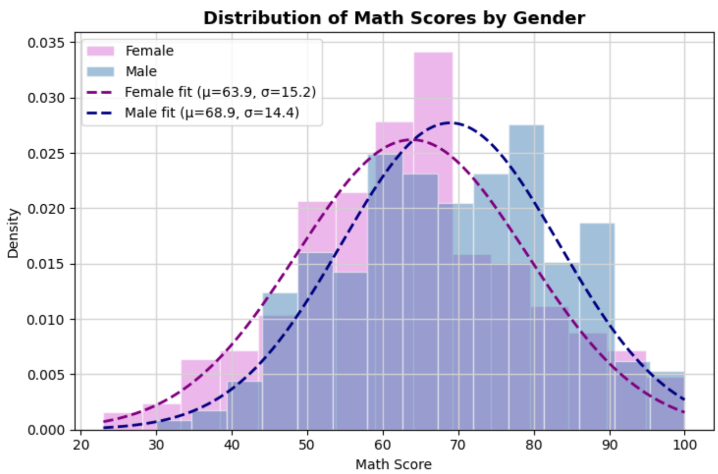

Figure 2. Boxplot of Mathematics scores by gender.

4.2.2 Language Scores

The Shapiro–Wilk test does not reject normality (males: $W = 0.991$, $p = 0.139$; females: $W = 0.990$, $p = 0.085$). Both distributions appear approximately normal, with female scores showing a slight right skew. Levene's test confirms equal variances ($W = 0.160$, $p = 0.690$).

The two-sample t-test yields $t = -6.409$ with 484 degrees of freedom and $p < 0.001$. We reject H_0, therefore female students score significantly higher than male students in Language. Figure 3 visualises this difference.

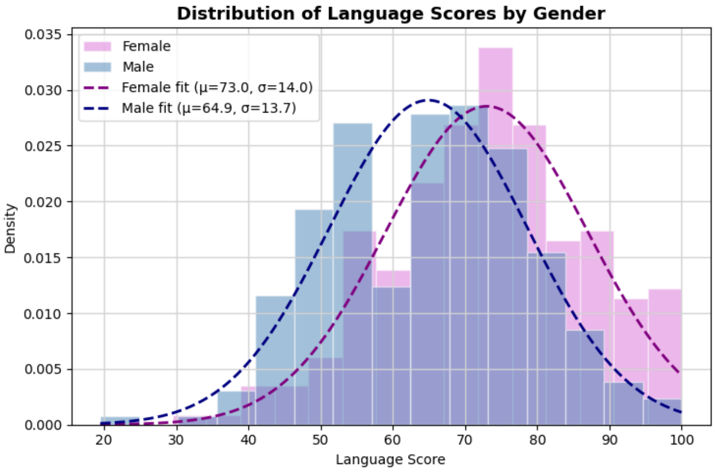

Figure 3. Boxplot of Language scores by gender.

4.3 Differences Across Parental Education Levels (Research Question 2)

4.3.1 Assumption Checks

The Shapiro–Wilk test rejects normality for two of the four Math groups (associate's degree: $W = 0.985$, $p = 0.049$; master's degree: $W = 0.956$, $p = 0.032$), while the remaining groups pass the test. For Language, all four groups satisfy the normality assumption ($p > 0.05$ in each case). Levene's test confirms equal variances for both Mathematics ($W = 1.275$, $p = 0.282$) and Language ($W = 0.169$, $p = 0.917$).

Since the normality assumption is violated in some Math subgroups, we report both the one-way ANOVA and the Kruskal–Wallis test.

4.3.2 ANOVA and Kruskal–Wallis Results

For Mathematics, the one-way ANOVA yields $F(3,482) = 6.223$, $p < 0.001$, and the Kruskal–Wallis test confirms this result ($H = 15.680$, $p = 0.001$). For Language, the ANOVA yields $F(3,482) = 14.241$, $p < 0.001$, with the Kruskal–Wallis test in agreement ($H = 36.708$, $p < 0.001$). In both subjects, we reject H_0: mean scores differ significantly across parental education levels.

4.3.3 Post-Hoc Pairwise Comparisons

Table 3 presents the adjusted p-values from both Tukey's HSD and Dunn's test with Bonferroni correction.

Table 3. Post-hoc pairwise comparisons across parental education levels. Adjusted p-values are shown; significant pairs ($p < 0.05$) are marked in bold.

		Math		Language	
Group 1	Group 2	p_{Tukey}	p_{Dunn}	p_{Tukey}	p_{Dunn}
Associate's	Bachelor's	0.988	1.000	0.660	1.000
Associate's	High school	**0.004**	**0.015**	$<$ **0.001**	$<$ **0.001**
Associate's	Master's	0.822	1.000	**0.043**	0.127
Bachelor's	High school	**0.008**	**0.018**	$<$ **0.001**	$<$ **0.001**
Bachelor's	Master's	0.951	1.000	0.437	1.000
High school	Master's	**0.006**	**0.012**	$<$ **0.001**	$<$ **0.001**

In both subjects, the high school group differs significantly from all other education levels under both tests. In Language, Tukey's HSD additionally detects a significant difference between the associate's degree and master's degree groups ($p = 0.043$), although Dunn's test does not confirm this after Bonferroni correction ($p = 0.127$). No significant differences are found among the associate's, bachelor's, and master's degree groups in Mathematics.

5 Summary

5.1 Summary of Findings

This report investigated two research questions: (1) whether Mathematics and Language scores differ significantly between male and female students, and (2) whether scores differ across parental education levels.

For Research Question 1, significant gender differences were found in both subjects. Male students scored significantly higher in Mathematics, while female students scored significantly higher in Language.

For Research Question 2, scores differ significantly across parental education levels in both subjects. Post-hoc comparisons showed that the high school group scored significantly lower than all other groups. No significant differences were found among the associate's, bachelor's, and master's degree groups in Mathematics. In Language, the results were mixed: Tukey's HSD detected a borderline difference between the associate's and master's degree groups, which Dunn's test did not confirm after Bonferroni correction. These findings are consistent with the research done by Muhtadi et al. (2023), who found that parental education level is positively associated with mathematics achievement.

5.2 Further Analysis

An important question that arose during the analysis was whether the observed gender effect could be an artefact of an imbalanced distribution of parental education levels between male and female students. To investigate this, we performed a stratified analysis: for each combination of parental education level and subject, we tested the gender difference separately, applying a Bonferroni correction across all eight comparisons ($\alpha_{\text{adj}} = 0.05/8 = 0.00625$). The results showed that the female advantage in Language remained significant in three of the four education groups (associate's, bachelor's, and high school), while the male advantage in Mathematics was significant only in the high school group after correction. This suggests that the Language effect is holds across education levels, whereas the Mathematics effect may be partly driven by the high school subgroup.

5.3 Limitations

Several limitations should be noted. The dataset does not include potential outside variables such as household income and school quality, which may partly explain the observed differences. The master's degree group is relatively small ($n = 59$) compared to others, which reduces the statistical power of comparisons involving this group.

References

Muhtadi, D. et al. (2023). “The Role of Students’ Beliefs, Parents’ Educational Level, and the Mediating Role of Attitude and Motivation in Students’ Mathematics Achievement”. In: *The Asia-Pacific Education Researcher*.